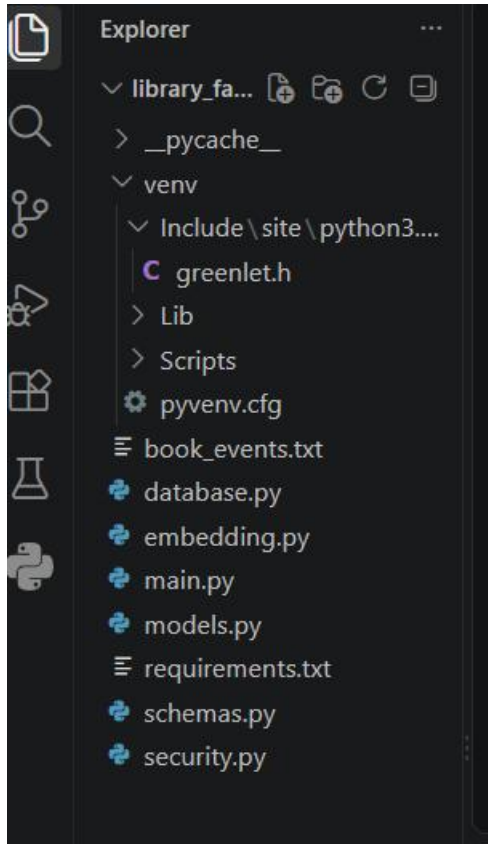


PRACTICAL WEEK-5

1. Project Structure



2. Installed Packages

```
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 6375, in startup
    await handler()
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\main.py", line 38, in startup
    async with engine.begin() as connection:
File "C:\Users\madhulika\AppData\Local\Programs\Python\Python311\Lib\contextlib.py", line 210, in __aenter__
    return await anext(self.gen)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\engine.py", line 1068, in begin
    async with conn:
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\base.py", line 121, in __aenter__
    return await self.start(is_ctxmanager=True)
```

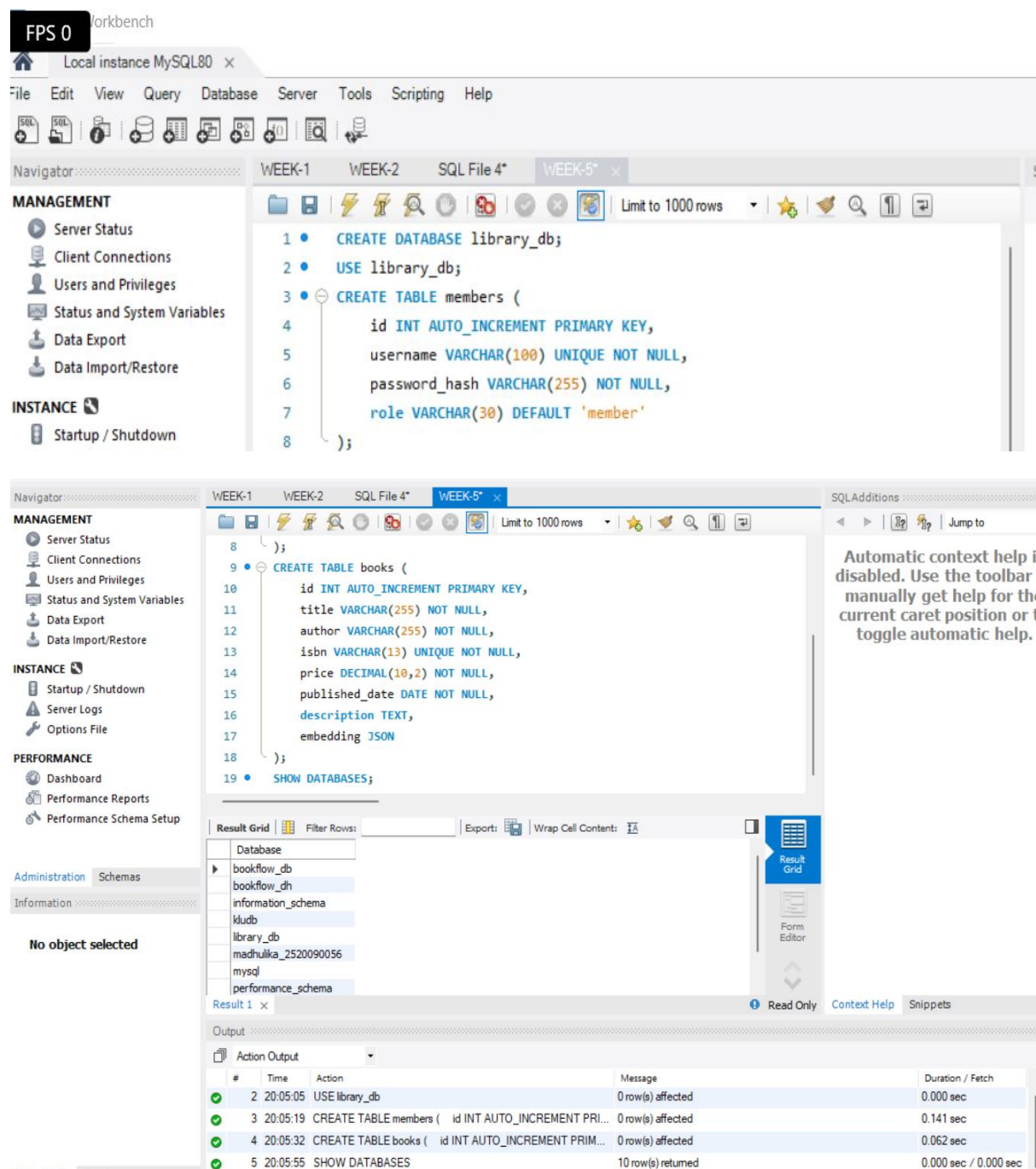
```
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 144, in __init__
    self._dbapi_connection = engine.raw_connection()
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 3319, in raw_connection
    return self.pool.connect()
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 448, in connect
    return _ConnectionFairy.checkout(self)
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _checkout
    fairy = _ConnectionRecord.checkout(pool)
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in checkout
    rec = pool._do_get()
    ~~~~~
```

```

File "asynmy/connection.pyx", line 976, in connect
File "asynmy/connection.pyx", line 954, in asynmy.connection.Connection.connect
File "asynmy/connection.pyx", line 1316, in _request_authentication
File "C:\Users\madhu\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\asynmy\auth.py", line 246, in caching_sha2_password_auth
    pkt = await _roundtrip(conn, data)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\madhu\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\asynmy\auth.py", line 116, in _roundtrip
    pkt = await conn.read_packet()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "asynmy/connection.pyx", line 1087, in read_packet
File "asynmy/protocol.pyx", line 956, in asynmy.protocol.MysqlPacket.raise_for_error
File "asynmy/protocol.pyx", line 957, in asynmy.protocol.MysqlPacket.raise_for_error
File "asynmy/errors.pyx", line 141, in asynmy.errors.raise_mysql_exception
File "asynmy/errors.pyx", line 156, in asynmy.errors.raise_mysql_exception
sqlalchemy.exc.OperationalError: (asynmy.errors.OperationalError) (1045, "Access denied for user 'root'@'localhost' (using password: YES)")

```

3. MySQL Setup



The screenshot shows a database management tool interface. On the left, there is a sidebar with categories: MANAGEMENT (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), INSTANCE (Startup / Shutdown, Server Logs, Options File), PERFORMANCE (Dashboard, Performance Reports, Performance Schema Setup), Administration (selected), and Schemas. Below Administration, it says "No object selected". The main area displays SQL code in a text editor with line numbers 10 to 21. The code defines a table structure and executes several commands. Below the editor, there is a "Result Grid" showing the output of the SQL commands. The "Output" pane at the bottom shows a log of actions and their messages.

SQL Code:

```

10 id INT AUTO_INCREMENT PRIMARY KEY,
11 title VARCHAR(255) NOT NULL,
12 author VARCHAR(255) NOT NULL,
13 isbn VARCHAR(13) UNIQUE NOT NULL,
14 price DECIMAL(10,2) NOT NULL,
15 published_date DATE NOT NULL,
16 description TEXT,
17 embedding JSON
18 );
19 SHOW DATABASES;
20 USE library_db;
21 SHOW TABLES;

```

Result Grid:

Tables_in_library_db
books
members

Output Log:

#	Time	Action	Message
4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected
5	20:05:55	SHOW DATABASES	10 row(s) returned
6	20:07:46	USE library_db	0 row(s) affected
7	20:07:52	SHOW TABLES	2 row(s) returned

4. Database.py

The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named "library_fastapi_project" with various files and folders. The code editor shows the content of "database.py".

Code in database.py:

```

1 from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker, create_async_engine
2 from sqlalchemy.orm import DeclarativeBase
3
4 DATABASE_URL = (
5     "mysql+asyncmy://root:Madhu%40248"
6     "@localhost:3306/library_db"
7 )
8
9 engine = create_async_engine(
10     DATABASE_URL,
11     echo=True
12 )
13
14 AsyncSessionLocal = async_sessionmaker(
15     engine,
16     expire_on_commit=False
17 )
18
19
20 class Base(DeclarativeBase):
21     pass
22
23
24 async def get_db():
25     async with AsyncSessionLocal() as session:
26         yield session

```

5. Embedding.py

```
library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3....
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── ...

embedding.py > text_to_vector
1  import hashlib
2  import math
3
4
5  VECTOR_DIM = 384
6
7
8  def text_to_vector(text: str):
9      vector = []
10
11      for i in range(VECTOR_DIM):
12          h = hashlib.sha256(
13              f"{text}-{i}".encode()
14          ).digest()
15
16          value = int.from_bytes(h[:4], "big") / (2**32 - 1)
17          vector.append(value)
18
19      norm = math.sqrt(
20          sum(x * x for x in vector)
21      )
22
23      if norm == 0:
24          return vector
25
26      return [x / norm for x in vector]
```

6. Main.py

```
library_fa...
├── __pycache__
├── venv
├── Include\site\python3....
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── ...

main.py > login
1  from fastapi import (
2      BackgroundTasks,
3      Depends,
4      FastAPI,
5      HTTPException,
6      Query,
7      status
8  )
9
10 from fastapi.security import OAuth2PasswordRequestForm
11
12 from sqlalchemy import select
13 from sqlalchemy.ext.asyncio import AsyncSession
14
15 from database import Base, engine, get_db
16 from embedding import text_to_vector
17 from models import Book, Member
18 from schemas import BookSchema, MemberRegister, TokenResponse
19
20 from security import (
21     create_access_token,
22     get_current_member,
23     hash_password,
24     verify_password
25 )
26
27 import math
28
29
30 app = FastAPI(
31     title="Library Backend & Security API",
32     version="1.0"
33 )
34
35
36 @app.on_event("startup")
```

7. Models.py

```
library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3...
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

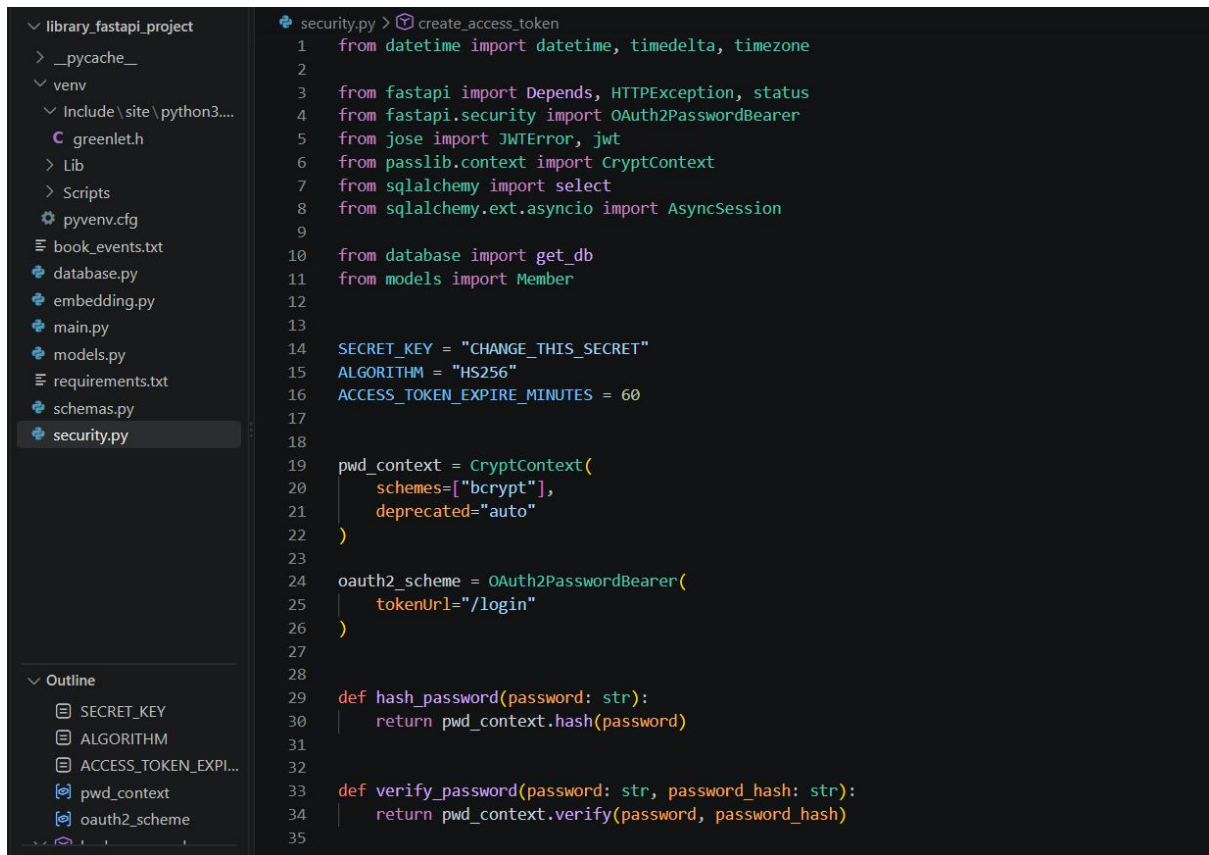
```
models.py > Book
1  from sqlalchemy import Column, Integer, String, Text, DECIMAL, Date, JSON
2  from database import Base
3
4
5  class Member(Base):
6      __tablename__ = "members"
7
8      id = Column(Integer, primary_key=True, index=True)
9      username = Column(String(100), unique=True, nullable=False)
10     password_hash = Column(String(255), nullable=False)
11     role = Column(String(30), default="member")
12
13
14     class Book(Base):
15         __tablename__ = "books"
16
17         id = Column(Integer, primary_key=True, index=True)
18         title = Column(String(255), nullable=False)
19         author = Column(String(255), nullable=False)
20         isbn = Column(String(13), unique=True, nullable=False)
21         price = Column(DECIMAL(10, 2), nullable=False)
22         published_date = Column(Date, nullable=False)
23         description = Column(Text)
24         embedding = Column(JSON)
```

8. Schemas.py

```
Explorer
├── library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3...
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

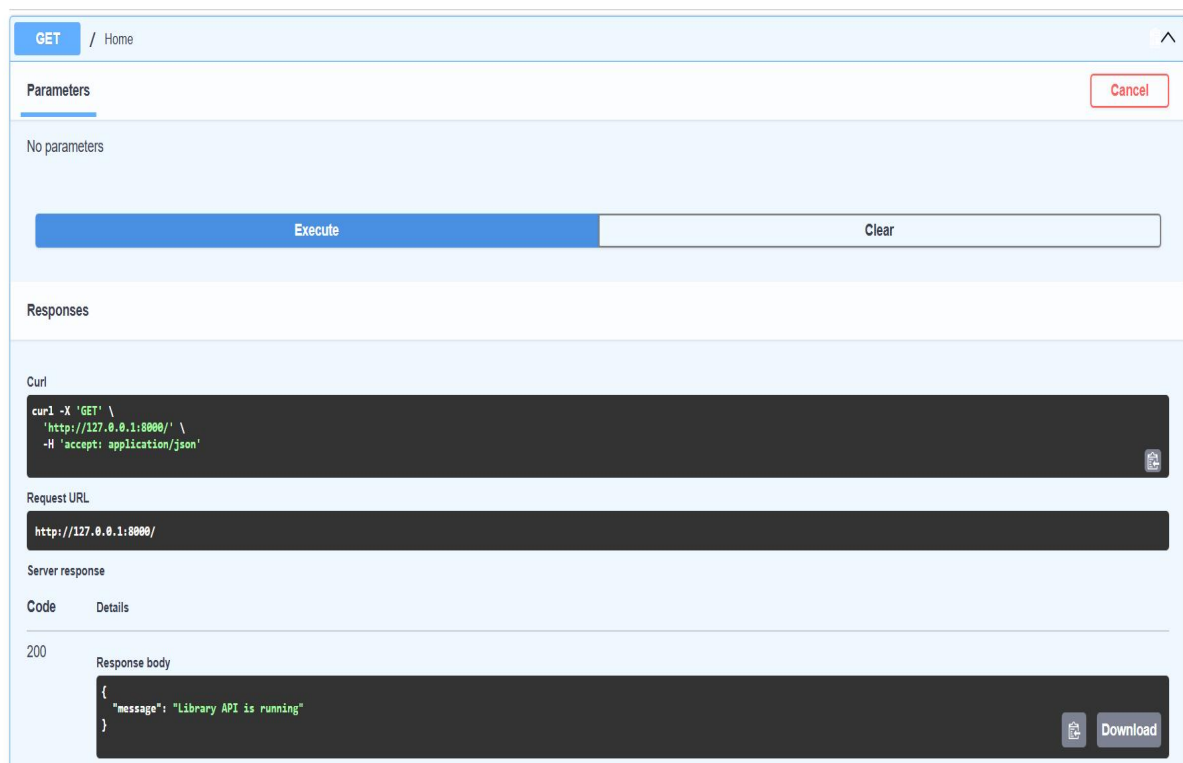
```
schemas.py > MemberRegister
1  from pydantic import BaseModel, Field, field_validator
2  from datetime import date
3
4
5  class BookSchema(BaseModel):
6      title: str
7      author: str
8      isbn: str = Field(min_length=13, max_length=13)
9      price: float = Field(gt=0)
10     published_date: date
11     description: str | None = None
12
13     @field_validator("published_date")
14     @classmethod
15     def validate_published_date(cls, value):
16         if value > date.today():
17             raise ValueError("published_date cannot be in the future")
18         return value
19
20
21     class MemberRegister(BaseModel):
22         username: str
23         password: str = Field(min_length=6)
24
25
26     class TokenResponse(BaseModel):
27         access_token: str
28         token_type: str
```

9. Security.py



```
security.py > create_access_token
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18
19 pwd_context = CryptContext(
20     schemes=["bcrypt"],
21     deprecated="auto"
22 )
23
24 oauth2_scheme = OAuth2PasswordBearer(
25     tokenUrl="/login"
26 )
27
28
29 def hash_password(password: str):
30     return pwd_context.hash(password)
31
32
33 def verify_password(password: str, password_hash: str):
34     return pwd_context.verify(password, password_hash)
35
36
```

10. Test Registration



GET / Home

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body <pre>{ "message": "Library API is running" }</pre>

Download

POST/registerRegister

Parameters

CancelReset

No parameters

Request bodyrequired

application/json

Edit ValueSchema

```
{
  "username": "Madhu",
  "password": "library123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Madhu",
    "password": "library123"
  }'
```

Request URL

http://127.0.0.1:8080/register

Server response

CodeDetails

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "Madhu",
  "role": "member"
}
```

Download

Response headers

```
content-length: 79
content-type: application/json
date: Mon, 14 Sep 2026 16:05:16 GMT
server: uvicorn
```

Responses

11. Test Login

POST/login Login

Parameters

CancelReset

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)

pattern: ^password\$

password

☐ Send empty value

username * required

string

Madhu

password * required

string(\$password)

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

Madhu

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/login' \
-H 'accept: application/json' \
-H 'Content-type: application/x-www-form-urlencoded' \
-d 'grant_type=password&username=Madhu&password=library123&scope=&client_id=Madhu&client_secret=library123'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code

Details

200

Response body

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVj9.eyJzdmI3OiJNYWModSIsInVzZGU4OiJtZWliZXI1LCJleHAiOjE3ODk0NDU5Nzh9.BSylTtpIyOmAoCo0hwV3PUg1xVb9I2uZ8d08yfy3Vx4",
  "token_type": "bearer"
}
```

Download

Response headers

```
content-length: 186
content-type: application/json
date: Mon, 14 Sep 2026 16:12:57 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header.

Example Value

Schema

12. Test POST/Books

POST

/books Create Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Python for Beginners",
  "author": "John Smith",
  "isbn": "123",
  "price": -100.00
  "published_date": "2024-06-10",
  "description": "An introduction to Python programming."
}
```

Execute

Clear

Responses

curl -X POST -H 'http://127.0.0.1:8000/books' \-H 'accept: application/json' \-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJlbnVudSIsInVubGUoiOiJtZWliZXIiLCJ1eHAiOiJES3B0dXQyZ9.Lq7z0UsTHMKMcQIBzYEgTX64tTGk1sHhu4JgNn8II0' \-H 'Content-Type: application/json' \-d '{\n \"title\": \"Python for Beginners\", \n \"author\": \"John Smith\", \n \"isbn\": \"123\", \n \"price\": 450.00, \n \"published_date\": \"2024-06-10\", \n \"description\": \"An introduction to Python programming.\"\n}'

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

422

Error: Unprocessable Entity

Response body

```
{\n  \"detail\": [\n    {\n      \"type\": \"string_too_short\", \n      \"loc\": [\n        \"body\", \n        \"isbn\" \n      ], \n      \"msg\": \"String should have at least 13 characters\", \n      \"input\": \"123\", \n      \"ctx\": {\n        \"min_length\": 13 \n      }\n    }\n  ]\n}
```

Response headers

```
content-length: 150\ncontent-type: application/json\ndate: Mon, 14 Sep 2026 16:28:41 GMT\nserver: uvicorn
```

GET
/books/search
Search Books

Parameters
Cancel

Name	Description
q * required string (query)	q = python minLength: 1
limit integer (query)	5 maximum: 20 minimum: 1

Execute
Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=q%20%3D%20python&limit=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=q%20%3D%20python&limit=5
```

Server response

13. book_event.txt

library_fastapi_project
__pycache__
venv
Include \site \python3....
greenlet.h
Lib
Scripts
pyvenv.cfg
book_events.txt
database.py
embedding.py
main.py
models.py
requirements.txt
schemas.py
security.py

book_events.txt
1 New Book Added: id=1, title=Python for Beginners
2

